

Machine Learning for Dynamic Bitcoin Allocation: A Risk-Adjusted, Long-Only Strategy

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Group 101 (G101) - Class T22 - 2025-2A - São Paulo **Business partner:** Hashdex - **Advisor:** Prof. [advisor name]

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Resumo

Investidores institucionais brasileiros enfrentam um dilema ao buscar exposição a Bitcoin (BTC): o ativo oferece prêmio de risco no longo prazo, mas com quedas (drawdowns) de 60% a 80% que são incompatíveis com a maioria dos mandatos. Este trabalho, desenvolvido com a gestora de criptoativos Hashdex, apresenta uma solução de aprendizado de máquina que decide semanalmente qual fração de uma carteira manter em BTC versus um indexador de caixa (o CDI), de forma **long-only** (0% a 100%, sem venda a descoberto) e otimizada para **retorno ajustado ao risco de queda** (Sortino). O núcleo é um ensemble duplo de árvores de decisão com gradient boosting (XGBoost) que prevê o retorno do BTC em três dias; a previsão é convertida em uma alocação dimensionada por um filtro de regime de mercado e pela confiança do modelo, e limitada por controles de risco automáticos (kill switch, redução por acurácia e monitor de drift). A validação seguiu práticas de finanças quantitativas: walk-forward out-of-sample com purga temporal, teste de permutação de rótulos, Sharpe deflacionado e validação com múltiplas sementes. Em um backtest de 4,28 anos out-of-sample, a estratégia obteve Sortino diário de 3,53 e drawdown máximo de -7,1%, contra -21,7% de uma alocação estática 30% BTC/70% CDI e -66,7% de uma posição comprada em BTC. O sinal é estatisticamente real (nenhuma de 100 permutações superou o baseline; $p < 0,01$). A solução encontra-se em fase de paper trade, com um gate de decisão de capital previsto para o terceiro trimestre de 2026.

Palavras-chave: aprendizado de máquina; alocação de ativos; bitcoin; gestão de risco; aprendizado de máquina em finanças.

Abstract

Brazilian institutional investors face a dilemma when seeking Bitcoin (BTC) exposure: the asset offers a long-run risk premium but with 60%-80% drawdowns that are incompatible with most mandates. This project, developed with the crypto asset manager Hashdex, presents a machine-learning solution that decides weekly what fraction of a portfolio to hold in BTC versus a cash benchmark (the Brazilian interbank rate, CDI), in a **long-only** manner (0% to 100%, no short selling) and optimized for **downside risk-adjusted return** (the Sortino ratio). The core is a dual ensemble of gradient-boosted decision trees (XGBoost) that forecasts the three-day forward BTC return; the forecast is converted into an allocation scaled by a

market-regime filter and the model's confidence, and bounded by automated risk controls (kill switch, accuracy de-risk, and a feature-drift monitor). Validation followed quantitative-finance best practice: walk-forward out-of-sample testing with temporal purging, label permutation testing, the Deflated Sharpe Ratio, and multi-seed validation. In a 4.28-year out-of-sample backtest the strategy achieved a daily Sortino ratio of 3.53 and a maximum drawdown of -7.1%, versus -21.7% for a static 30% BTC / 70% CDI allocation and -66.7% for a buy-and-hold BTC position. The signal is statistically real (none of 100 label permutations beat the baseline; $p < 0.01$). The solution is in a paper-trading phase, with a capital-allocation decision gate planned for the third quarter of 2026.

Keywords: machine learning; asset allocation; bitcoin; risk management; financial machine learning.

1. Introduction

1.1 Partner company context

Hashdex is an asset manager specialized in crypto assets, operating index products and investment vehicles for institutional and retail investors. The relevant area for this project is the quantitative/portfolio-management desk, responsible for deciding how much exposure to risky crypto assets a given vehicle should carry over time.

1.2 Motivation

Bitcoin allocation decisions are frequently **discretionary** - hard to audit, to reproduce, and to scale across vehicles or review by a risk committee. A systematic, auditable process that captures most of Bitcoin's upside while sharply limiting its drawdown would be strategically valuable: it reduces key-person risk, standardizes a recurring decision, and is defensible to investors and regulators.

1.3 Problem definition

The core problem is to **capture Bitcoin's risk premium with a fraction of its drawdown**, systematically. The baseline (the current alternative) is a static allocation. Over the 2022-2026 test window, in BRL: a 100% BTC buy-and-hold position returned +15.5% CAGR but suffered a **-66.7% maximum drawdown**; a static 30% BTC / 70% CDI blend returned +17.0% CAGR with a -21.7% drawdown and a Sortino ratio of 1.13; cash (CDI) returned +13.0% with no drawdown. The opportunity is to dominate these baselines on a downside-risk-adjusted basis.

1.4 Proposed solution and expected contribution

We propose a computational pipeline that, each week, generates a target BTC/CDI allocation from a machine-learning forecast, scaled by market regime and model confidence and bounded by automated risk controls. The **contribution objective** is a strategy that, out-of-sample, materially improves the Sortino ratio and reduces maximum drawdown relative to static allocations, while remaining long-only.

1.5 Business objectives

For the partner, the expected results are: (i) a systematic, auditable allocation process that replaces discretionary judgment; (ii) superior downside risk-adjusted performance (target Sortino ≥ 2.5 ; conservative floor ≥ 1.5 after deflation); (iii) maximum drawdown materially below buy-and-hold; and (iv) an operationally reliable signal validated in paper trading before any capital is committed.

1.6 Structure of this report

Section 2 develops the solution: the applied rationale (2.1), the specification and development of the pipeline (2.2), and the assessment of impact and contribution to the business (2.3). Section 3 concludes and outlines next steps.

2. Solution development

2.1 Applied rationale

Business-area rationale. In crypto asset management, the dominant risk is the asymmetry of Bitcoin's return distribution - large, infrequent gains alongside deep, prolonged drawdowns. Market best practice for managing such exposure ranges from static rebalancing to discretionary market timing; both leave risk-adjusted return on the table. A systematic, regime-aware allocation that prioritizes downside protection is the gap this project addresses.

Technological rationale. A fixed rule (e.g., a permanent 30% BTC sleeve, or a pure moving-average crossover) does not adapt to regime and cannot size a position by the strength and confidence of a view. A supervised machine-learning model can. We adopt **gradient-boosted decision trees (XGBoost)** because, on tabular data with roughly 30 engineered features and a few thousand observations, gradient boosting empirically outperforms recurrent neural networks, random forests, and stacked meta-learners - all of which were tested and rejected on the target objective. Bagging many trees per ensemble reduces seed-to-seed variance and stabilizes the live signal.

Foundations from financial-ML methodology. The design incorporates established literature: the Sortino ratio (Sortino and Price, 1994) as an objective that penalizes only downside volatility - appropriate because Bitcoin's fat tails are predominantly on the upside; walk-forward validation with temporal **purging/embargo** and fractional differentiation from López de Prado (2018); the **Deflated Sharpe Ratio** (Bailey and López de Prado, 2014) to discount multiple-testing; and fractional-Kelly position sizing for the confidence scaling.

Design-choice rationale. The key operating parameters follow from principles, not parameter sweeps. The portfolio rebalances **weekly** because the decision frequency should match the rate at which the underlying signals (macro liquidity, on-chain data, trend regime) actually evolve - daily rebalancing would trade on noise and pay cost, while monthly would be too slow for crypto's fast regime shifts; rebalancing on **Friday** concentrates a full week of information into one auditable decision and makes weekend exposure (crypto trades 24/7) a deliberate choice. The model forecasts a short, **three-day** horizon - where return predictability is strongest - but **holds the position through the week**, because no materially better information arrives intra-week to justify the cost of trading; an **emergency rebalance** overrides this only on rare, regime-defining single-day moves (fat-tail events that are signal, not the noise the weekly cadence absorbs). The strategy is **long-only** because shorting a positively-drifting, fat-right-tailed asset is asymmetrically risky; positions are sized by model **confidence** (a fractional-Kelly intuition); and the regime multipliers are deliberately **conservative**, trading some backtest return for robustness to prediction and regime-classification error in live operation.

2.2 Specification and development

Data. The pipeline ingests roughly a dozen public and licensed sources spanning three families - macroeconomic (e.g., monetary aggregates, policy-rate proxies), on-chain (network and valuation metrics), and market/technical (price, volume, derivatives basis and funding). From these it derives about thirty

features. Feature construction is strictly **backward-looking**, a property enforced by an automated property test that perturbs the most recent observation and verifies that no past feature value changes - guarding against look-ahead bias.

Model and sizing. Two bagged XGBoost ensembles are trained: a regressor for the three-day forward BTC return and a classifier for its direction. The regression forecast is converted into a target allocation that is **scaled by two factors**: a market-regime filter (derived from long- and short-term moving averages) that reduces exposure in bearish regimes, and a **confidence weight** from the directional classifier, so the strategy bets larger only when the model is more certain. The result is clipped to the [0%, 100%] long-only range. The portfolio rebalances weekly, with an extra emergency rebalance on very large single-day moves, and the model is retrained on a fixed semi-annual schedule using an expanding window.

Risk controls. Three automated controls are applied to every signal and are part of the production code: a **kill switch** that caps exposure when cumulative drawdown breaches a threshold; an **accuracy de-risk** that halves exposure when rolling predictive accuracy deteriorates while the model remains overconfident; and a **population-stability monitor** that flags feature-distribution drift.

Validation methodology. The system is evaluated with walk-forward, out-of-sample testing (expanding window, temporal purge/embargo), multi-seed runs to quantify metric variance, label-permutation testing to establish a null distribution, and the Deflated Sharpe Ratio to discount the number of configurations tried.

Engineering. The solution is a one-command production pipeline with pinned dependencies for reproducibility, a configuration fingerprint that detects a stale model and forces retraining, and a test suite of 132 tests (covering look-ahead, determinism, train/serve parity, and the risk controls).

2.3 Assessment of impact and contribution to the business

All figures below are **out-of-sample backtest** results over 4.28 years (248 weekly rebalances), reported on a multi-seed basis and aware of realistic transaction costs (a few basis points per rebalance; results remain strong even under a pessimistic 50 bps assumption).

Table 1 - Risk-adjusted performance (4.28-year out-of-sample backtest, BRL).

Strategy	CAGR	Sortino (daily)	Max drawdown
Cash benchmark (CDI)	+13.0%	-	0%
Static 30% BTC / 70% CDI	+17.0%	1.13	-21.7%
100% Bitcoin (buy-and-hold)	+15.5%	0.56	-66.7%
This strategy (long-only ML)	+57.3%	3.53	-7.14%

The strategy delivered markedly higher downside-risk-adjusted performance than either static allocation, holding maximum drawdown to roughly **one-tenth** of buy-and-hold - a direct consequence of spending most of the time (~58%) in cash and concentrating exposure (~15% average) in favorable regimes.

Statistical significance. In label-permutation testing, **none of 100 shuffled-target runs beat the baseline** ($p < 0.01$). A feature-composition probe shows that signal + regime + confidence alone yield a Sortino of

2.19, while the ML magnitude forecast adds **+3.4** - i.e., the model is the source of the incremental edge, not a fixed rule. After applying a Deflated Sharpe Ratio for multiple-testing, the strategy still passes under a realistic estimate of the number of effective trials.

Early live (paper) evidence. In a strict out-of-sample window of the most recent year-to-date (105 days, 17 rebalances), the strategy returned **+19.67%** while Bitcoin buy-and-hold returned **-14.36%** and cash **+4.09%** - early corroboration, though concentrated in a small number of rebalances.

Business impact. For an illustrative (fictitious) allocated capital, the value created is the incremental return over the static baseline; because the operating cost of the solution is negligible relative to that incremental return, the decisive variable for return on investment is not cost but **whether the edge persists live** - precisely what the paper-trading phase and the decision gate exist to answer. The qualitative contribution is a systematic, auditable allocation process that reduces operational and key-person risk.

3. Conclusion

This project demonstrates that a disciplined machine-learning approach can turn a discretionary "how much Bitcoin to hold" decision into a **systematic, auditable, and risk-controlled** process that, in out-of-sample testing, substantially improves downside risk-adjusted returns and sharply reduces drawdown relative to static allocations. The evidence that the signal is statistically real - surviving permutation and deflation tests - distinguishes it from an overfit backtest.

We are deliberately conservative about the results. After discounting for multiple-testing, the realistic forward expectation is a Sortino in the 1.5-2.5 range and a drawdown of -15% to -25% - still superior to the static baselines, but well below the raw backtest. The return is also concentrated in a few decisive weeks, and the model depends on periodic retraining, which is why automated risk controls exist.

Next steps: complete the paper-trading window and pass the decision gate before committing capital; add model-explainability tooling (feature attribution) to the production signal; and, conditional on the gate, invest in production hardening and extend the same framework to additional asset/cash pairs. The model uses only market and on-chain data (no personal data), its computational footprint is low (CPU-only, retrained twice a year), and the project adopts an explicit honesty policy: results are always labeled as backtest/paper-trade and reported with their deflated, conservative range.

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